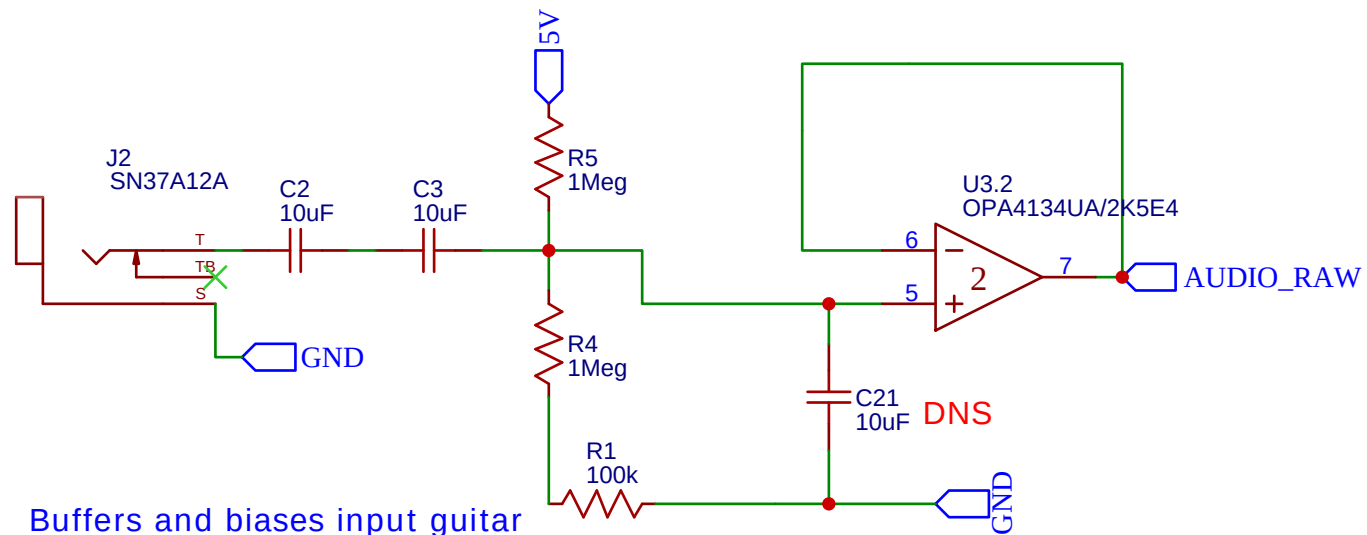
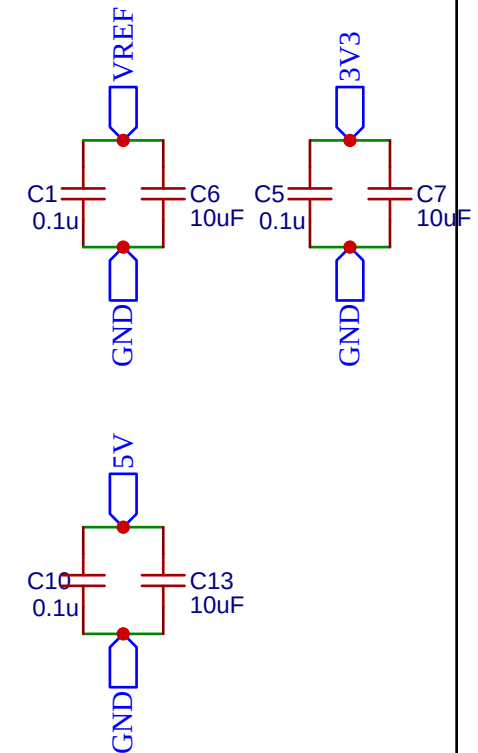
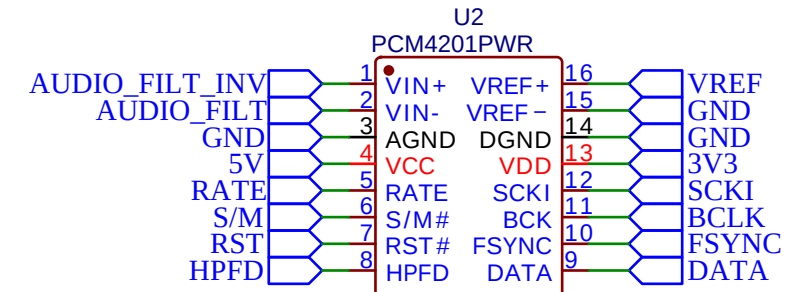


Audio Jack (1/4)

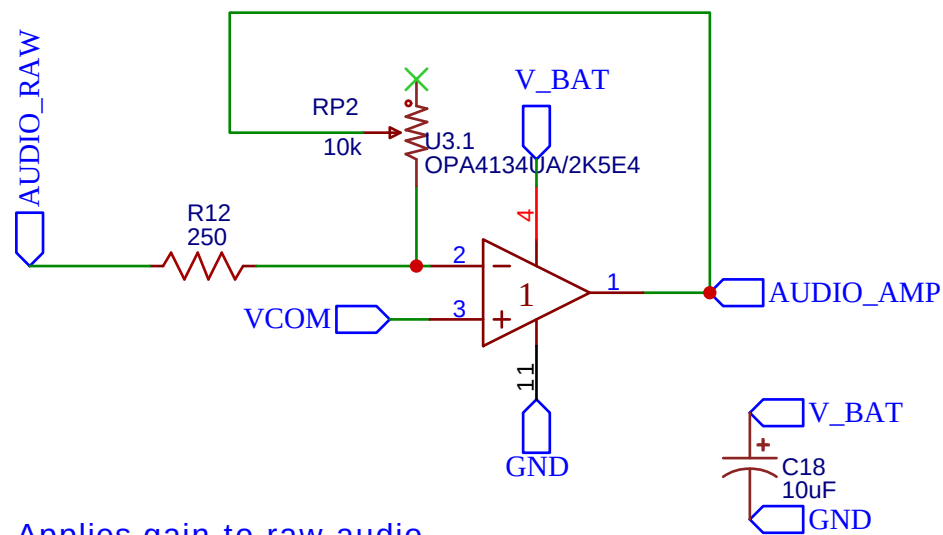


Buffers and biases input guitar signal to ~2.6V. Bias point chosen due to common mode considerations. Cap values chosen to provide good DC attenuation and little attenuation of 40hz+ frequencies.

A/D Converter

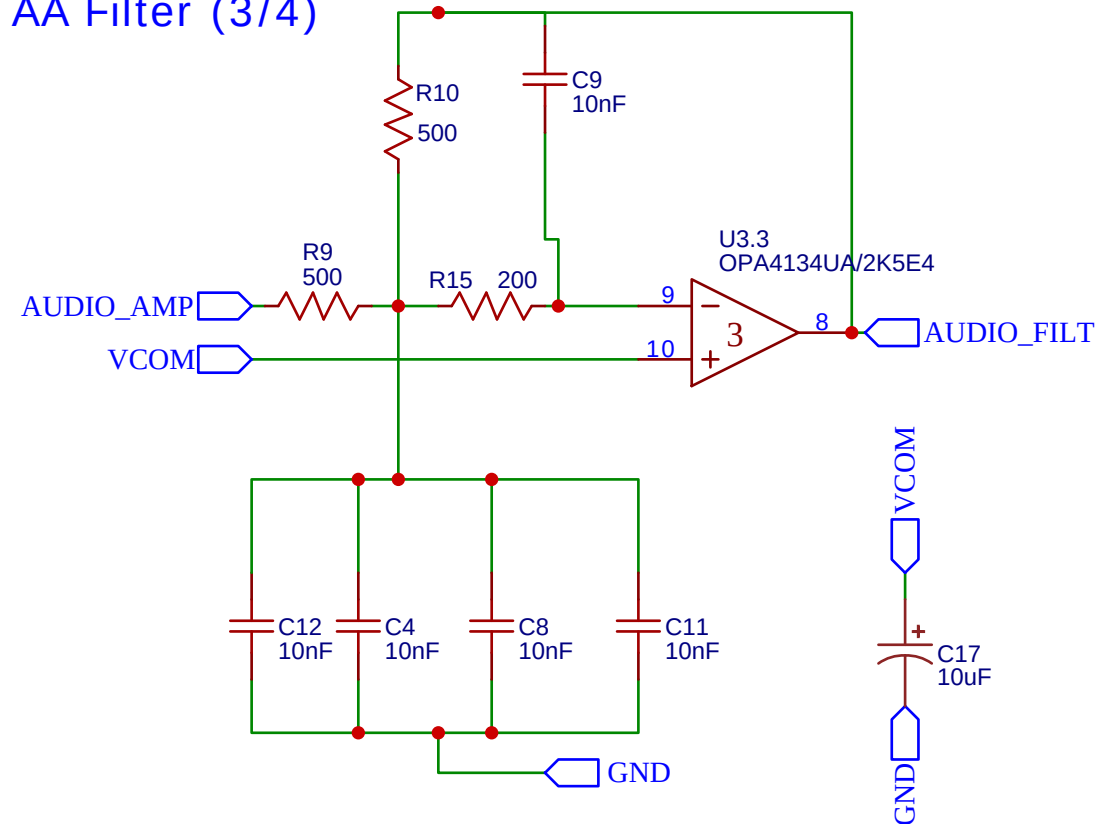


Gain Stage (2/4)

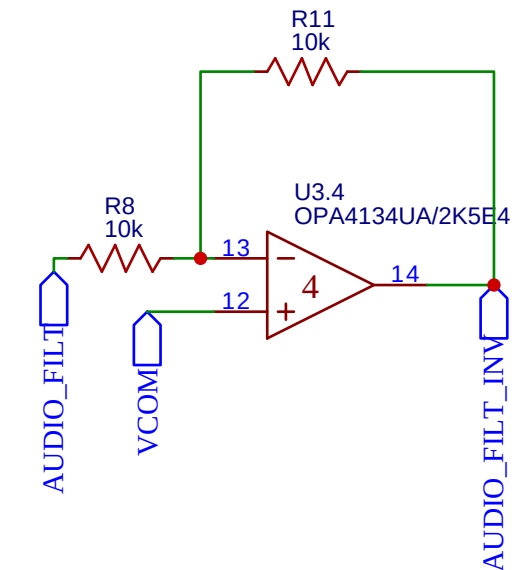


Applies gain to raw audio signal. Max gain is $10k/250 = 40$. Potentiometer can be adjusted to selected gain.

AA Filter (3/4)



Inverting Stage (4/4)



TITLE: Sheet_1		REV: 1.0
EasyEDA	Company: Your Company	Sheet: 1/1
	Date: 2025-05-15	Drawn By: rahulahoop

Power Supply

V_BAT 5V

GND V_{bat}

2-PIN JST Socket U10

U11 LD1117S50 CTR

GND VOUT VOUT VIN

C16 10uF

C15 100nF

GND GND GND

VCOM

The diagram shows a circuit for biasing the VCOM pin. A 5V supply is connected to a 1Meg resistor (R14), which is connected to a central node. This central node is also connected to a 1Meg resistor (R13) leading to a GND source, and a 100k resistor (R7) leading to another GND source. A 0.1uF capacitor (C20) is connected between the central node and a GND source. The central node is labeled VCOM.

The diagram shows the ESP32 pin configuration for the ESP32-DEVKITC. The pins are numbered 1 to 38. The connections are as follows:

- Pin 1: 3V3
- Pin 2: EN
- Pin 3: VP
- Pin 4: VN
- Pin 5: IO34
- Pin 6: IO35
- Pin 7: IO32
- Pin 8: IO33
- Pin 9: IO25
- Pin 10: IO26
- Pin 11: IO27
- Pin 12: DATA
- Pin 13: BCLK
- Pin 14: S/M
- Pin 15: GND
- Pin 16: GND
- Pin 17: D2
- Pin 18: D3
- Pin 19: CMD
- Pin 20: 5V
- Pin 21: CLK
- Pin 22: D0
- Pin 23: D1
- Pin 24: IO15
- Pin 25: IO16
- Pin 26: IO17
- Pin 27: IO18
- Pin 28: IO19
- Pin 29: HPFD
- Pin 30: RST
- Pin 31: FSXNC
- Pin 32: WIFI_CONN
- Pin 33: IO0
- Pin 34: IO1
- Pin 35: IO2
- Pin 36: IO3
- Pin 37: IO4
- Pin 38: IO5

Additional components and connections:

- R16: 50 Ohm resistor connected to RATE and SCKI pins.
- C14: 1mF capacitor connected to 3V3 and GND.
- C19: 1mF capacitor connected to 5V and GND.
- WIFI_CONN: Connected to pin 32.
- R2: 50 Ohm resistor connected to WIFI_CONN and LED1.
- LED1: LED RED_5MM connected to R2 and GND.

The following pins have been verified to work as I2S TX:

- GPIO 12
- GPIO 14
- GPIO 27

GPIO 32 is part of ADC Unit 1, which works during Wifi communication

TITLE: Sheet_2		REV: 1.0
	Company: Your Company	Sheet: 1/1
	Date: 2025-06-21 Drawn By: rahulahoop	